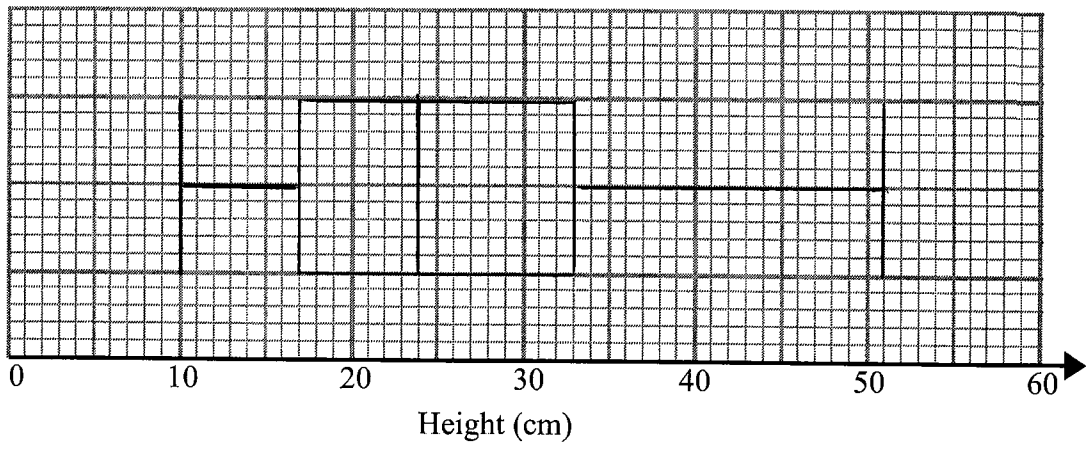
1 The table shows some information about the heights, in cm, of some plants.

Minimum	Lower Quartile	Median	Upper Quartile	Maximum
10	17	24	33	51

Draw a box plot for this information.



(Total for Question 1 is 2 marks)

2 Expand and Simplify $(x-2)(x+3)(x-4)$

$(x^2 + 3x - 2x - 6)(x - 4)$

$(x^2 + x - 6)(x - 4)$

$x^3 - 4x^2 + x^2 - 4x - 6x + 24$

$x^3 - 3x^2 - 10x + 24$

$x^3 - 3x^2 - 10x + 24$

(Total for Question 2 is 3 marks)

$3\overline{)1071}$ = 357

3 Write $3.2\dot{4}\dot{5}$ as a fraction in its simplest form.

$3.2\dot{4}\dot{5} = x$

$32.\dot{4}\dot{5} = 10x$

$3245.\dot{4}\dot{5} = 1000x$

$3213 = 990x$

$x = \frac{3213}{990}$

$\frac{3213}{990} = \frac{1071}{330} = \frac{357}{110}$

$\frac{357}{110}$

(Total for Question 3 is 3 marks)

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4 Find the equation of the line perpendicular to $5y - 4x = 6$ which passes through (0,7)

$5y = 4x + 6$

$y = \frac{4}{5}x + \frac{6}{5}$

$m = \frac{4}{5}$

perp $m = -\frac{5}{4}$

$y = -\frac{5}{4}x + 7$

(Total for Question 4 is 2 marks)

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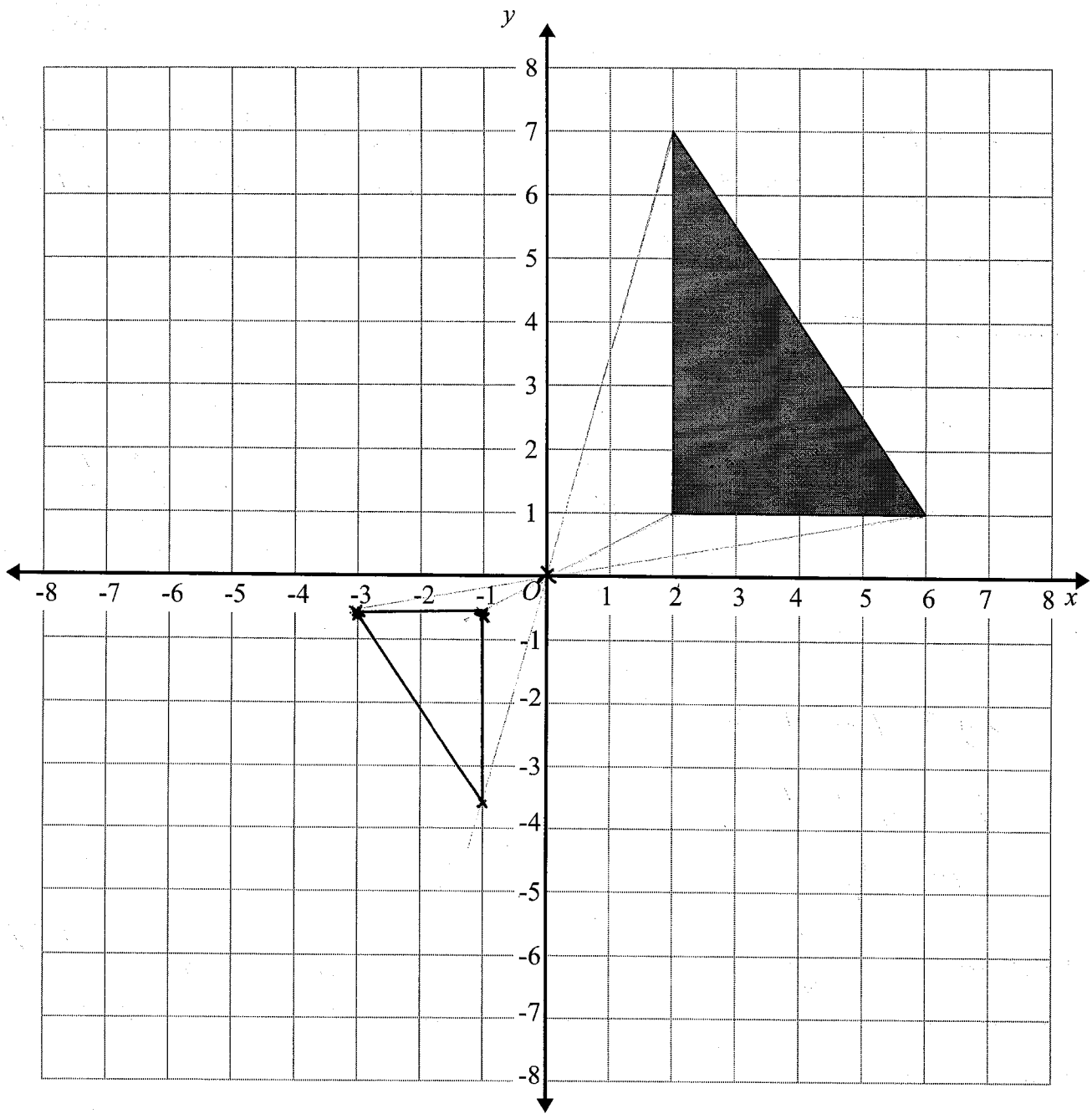
5 Find the value of $(27x^9)^{\frac{2}{3}}$

$(3x^3)^2$

$9x^6$

(Total for Question 5 is 2 marks)

6



On the grid, enlarge the triangle by scale factor -0.5. centre *O*.

(Total for Question 6 is 2 marks)

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \times -0.5 = \begin{pmatrix} -1 \\ -0.5 \end{pmatrix}$$

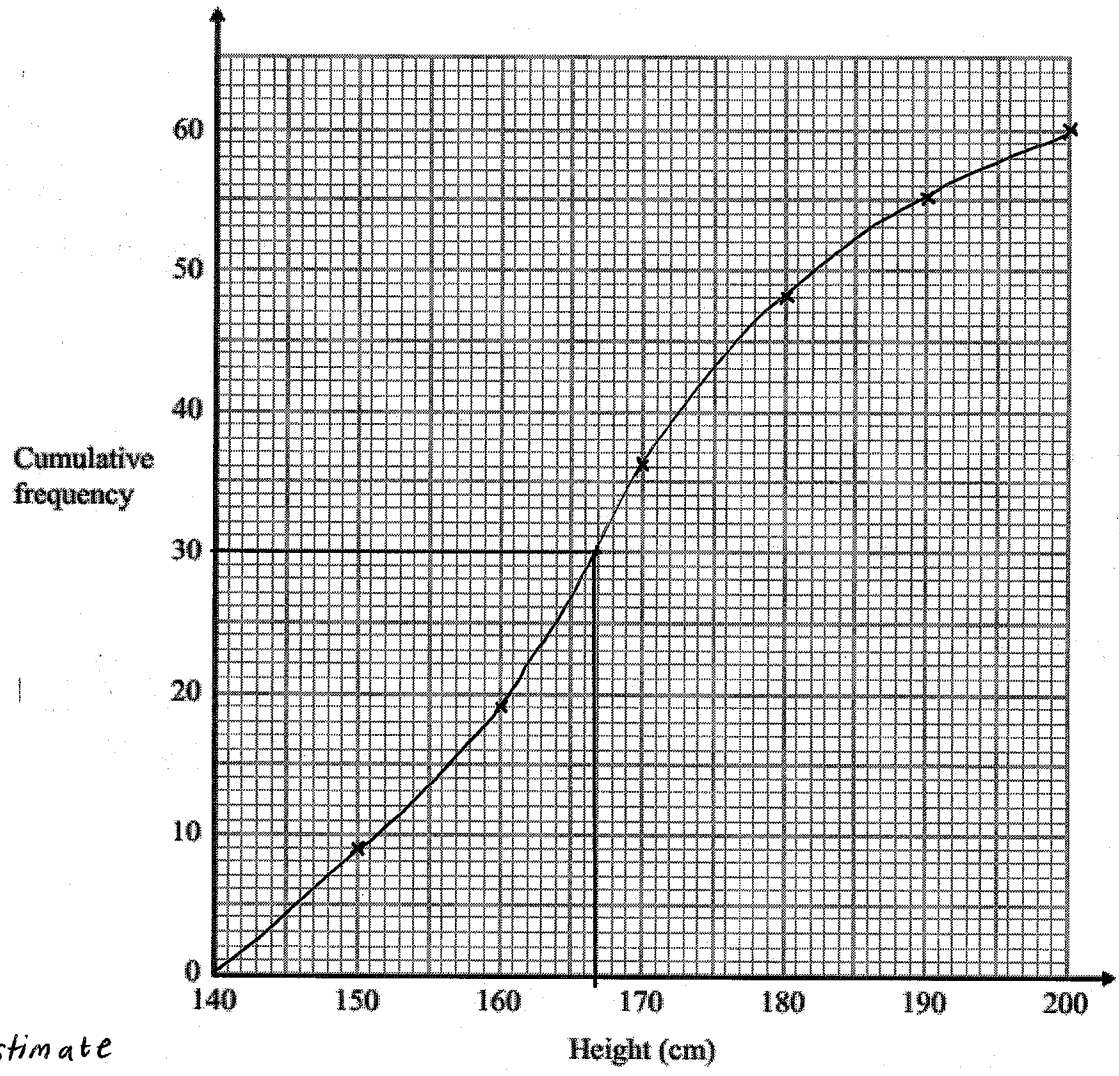
$$\begin{pmatrix} 6 \\ 1 \end{pmatrix} \times -0.5 = \begin{pmatrix} -3 \\ -0.5 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 7 \end{pmatrix} \times -0.5 = \begin{pmatrix} -1 \\ -3.5 \end{pmatrix}$$

7 The cumulative frequency table shows the height, in cm, of some tomato plants.

Height	Cumulative Frequency
$140 < h \leqslant 150$	9
$140 < h \leqslant 160$	19
$140 < h \leqslant 170$	36
$140 < h \leqslant 180$	48
$140 < h \leqslant 190$	55
$140 < h \leqslant 200$	60

(a) On the grid, plot a cumulative frequency graph for this information.



Estimate

(2)

(b) ~~Find~~ the median height.

.....167.....cm

166/167 (1)

(Total for Question 7 is 3 marks)

8 Make x the subject of the formula $\frac{a}{b} = \frac{x}{x-5}$

$a(x-5) = bx$

$ax - 5a = bx$

$ax - bx = 5a$

$x(a-b) = 5a$

$x = \frac{5a}{a-b}$

$\boxed{\text{or } x = \frac{-5a}{b-a}}$

$x = \frac{5a}{a-b}$

(Total for Question 8 is 3 marks)

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9 Write $(4 - \sqrt{3})^2$ in the form $a + b\sqrt{3}$, where a and b are integers.

$(4-\sqrt{3})(4-\sqrt{3})$

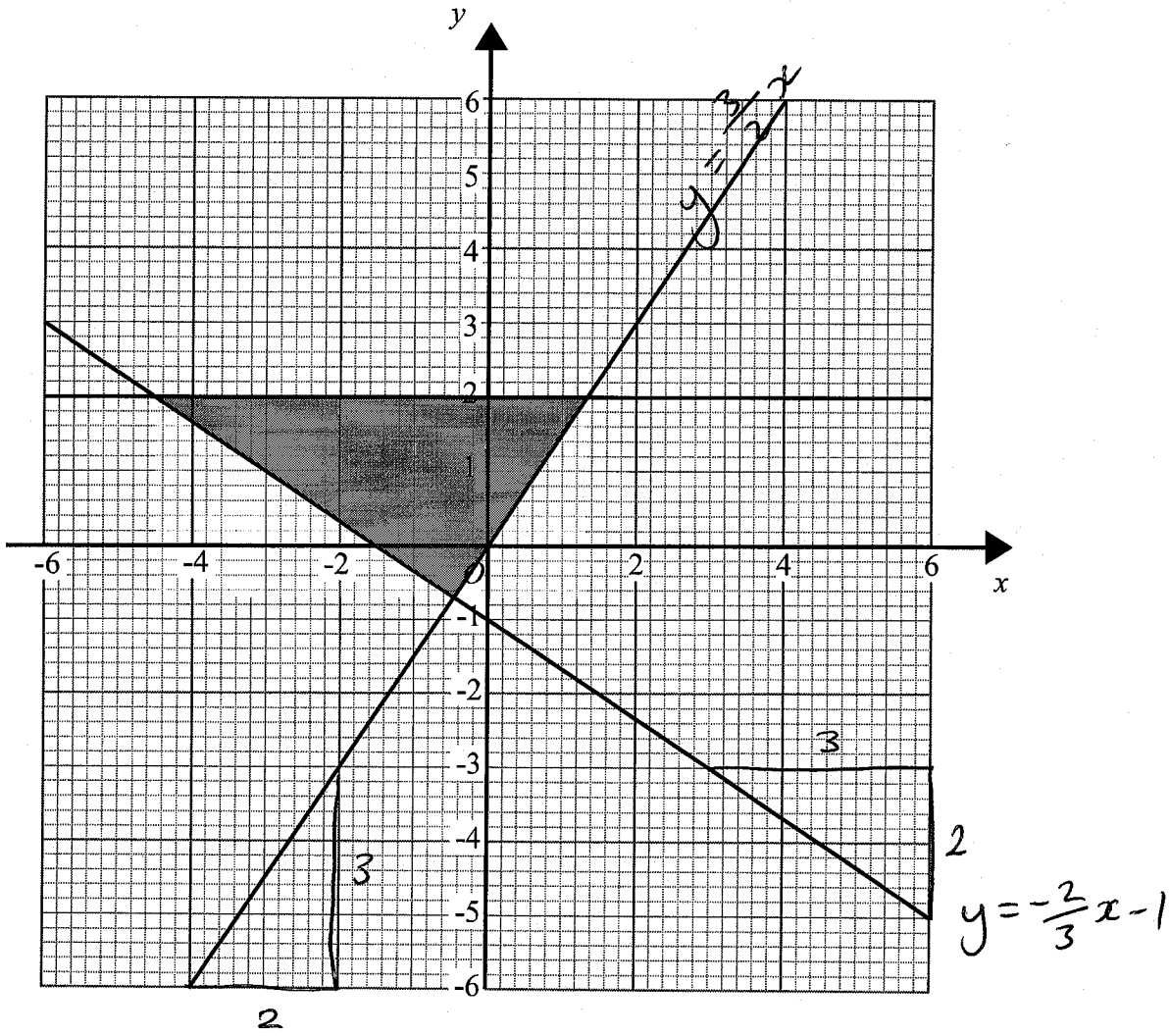
$16 - 4\sqrt{3} - 4\sqrt{3} + 3$

$19 - 8\sqrt{3}$

$19 - 8\sqrt{3}$

(Total for Question 9 is 2 marks)

10



Write down the three inequalities that define the shaded region

$y \leqslant 2$

$y \geqslant \frac{3}{2}x$

$y \geqslant -\frac{2}{3}x - 1$

(Total for Question 10 is 3 marks)